

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B. Tech CSE**

Semester: VI

Name of the Course: Security and Auditing

Course Code: TCS695

Time: 90 Minutes

Maximum Marks: 50

Note:

- i. Answer all the questions by choosing any one of the sub questions.
- ii. Each questions carries 10 marks.

Q1		(10 Marks)	CO1
(a)	Explain the following types of cyber threats. Social engineering attacks and malware attacks.		
	OR		
(b)	Demonstrate the working procedure of advanced persistent threats (APT) using a figure.		
Q2		(10 Marks)	CO3
(a)	Describe the following types of malware attacks. Spyware, ransomware and rootkit.		
	OR		
(b)	What is a security patch? Explain the following types of security patches. Hotfixes and cumulative updates.		
Q3		(10 Marks)	CO2
(a)	Describe the following types of hacker's tools. Reconnaissance tools and vulnerability scanners.		
	OR		
(b)	What is security governance? What are the key principles of security governance?		
Q4		(10 Marks)	CO5
(a)	Describe the implementation tiers of the NIST Cybersecurity Framework. How do they help organizations to improve their security posture?		
	OR		
(b)	Describe the responsibilities of the following roles under information security roles and positions. Security architect, compliance officer and incident responder.		
Q5		(10 Marks)	CO6
(a)	Discuss the punishment provisions under IT act 2000 for the following malicious activities. "Unauthorized access and hacking," "cyber stalking," "identity theft," and "publishing obscene material online."		

	OR	
(b)	Analyze the advantages and limitations of proxy firewalls compared to packet-filtering firewalls. Under what scenarios would a proxy firewall be the best choice?	